

Winter Crossword

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ONE ACROSS

Unfortunately, Santa's sleigh has broken down this year! Desiring to burn off the cookies and milk, he decides to walk between towns instead! For the county of Mathexford, there are 4 different towns. Each pair of towns has a singular road between them, which is blocked by snow with probability $\frac{1}{2}$ independent of any other roads. Given that Santa can deliver presents to children in every town with probability $\frac{p}{q}$, where p and q are relatively prime integers, find $q - p$.

ONE DOWN

Instead of spending time with family, Mister K opts to do math on Christmas Day. He has recently become fascinated with snowflakes and wants to make his own. Starting with an equilateral triangle, he replaces the middle third of each segment with two sides to form an equiangular bump at each iteration. Let K_n be the area of his snowflake after n iterations, and let

$$K_\infty = \lim_{n \rightarrow \infty} K_n.$$

Given that $\frac{K_3}{K_\infty}$ can be written in lowest terms as $\frac{p}{q}$, find $p + q$.

TWO DOWN

A baker is preparing a batch of special MathEX Christmas cookies, all of which are cut into the shape of the same irregular pentagon. The original recipe specifies the lengths of all five sides of the cookie. However, due to a spill in the kitchen, the measurement for the fifth side has been obscured.

The four visible side lengths are 5, 7, 12, and 15.

Assuming each cookie forms a non-degenerate pentagon (a polygon with non-zero area) and the length of the missing side is an integer, what is the maximum possible length of this fifth side?

THREE ACROSS

Unfortunately, the mayor of Candy Cane Lane has noticed that some holiday decorations look a bit too uniform. To keep the street festive and interesting, she introduces a rule for the Christmas lights. Each house on Candy Cane Lane has a string of lights that is either red, green, or blue. There are 6 houses in a row. To avoid monotony, the mayor requires that no three consecutive houses may have the same color of lights. If n is the number of different valid ways there are to decorate the street, find the product of the digits of $n - 4$.